

Homework 4 With Hints

1. Let $\mathcal{E} = (1, x, x^2)$ be the standard ordered basis of $(\mathbb{Z}_5)_2[x]$ and let $\mathcal{B} = (\vec{w}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \vec{w}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \vec{w}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix})$ be the ordered basis of the linear space given by

$$W = \left\{ \begin{pmatrix} w \\ x \\ y \\ z \end{pmatrix} \in \mathbb{Z}_5^4 : z = w + x + y \right\}$$

Note: you do not have to prove that W is a subspace of $\mathbb{Z}_5^4$ nor that $\mathcal{B}$ is an ordered basis of W .

- (a) Let $S : (\mathbb{Z}_5)_2[x] \rightarrow W$ be a linear transformation. Suppose that

$$[S]_{\mathcal{B}}^{\mathcal{E}} = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 4 & 4 \\ 2 & 1 & 3 \end{pmatrix}$$

Compute $S(a + bx + cx^2)$ for every $a + bx + cx^2 \in (\mathbb{Z}_5)_2[x]$.

- (b) Consider the linear transformation $T : (\mathbb{Z}_5)_2[x] \rightarrow W$ defined by

$$T(a + bx + cx^2) = \begin{pmatrix} a + 4b \\ c + b + 3a \\ 2a + c \\ a + 2c \end{pmatrix} \quad \forall a + bx + cx^2 \in (\mathbb{Z}_5)_2[x]$$

Compute the representation matrix $[T]_{\mathcal{B}}^{\mathcal{E}}$.

- (c) Prove that T is invertible and compute the representation matrix $[T^{-1}]_{\mathcal{E}}^{\mathcal{B}}$.

Hint. Use the inversion algorithm.

2. Let V be a vector space over $\mathbb{R}$ with $\dim(V) = 3$. Let $\mathcal{A} = (v_1, v_2, v_3)$ and $\mathcal{B} = (w_1, w_2, w_3)$ be two ordered bases of V and let $v, w \in V$. Suppose that

$$[id_V]_{\mathcal{B}}^{\mathcal{A}} = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

- (a) Suppose that $[w]_{\mathcal{B}} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$. Compute $[w]_{\mathcal{A}}$.

- (b) Suppose that $[v]_{\mathcal{B}} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$. Compute $[v + w]_{\mathcal{A}}$.

3. Let V be a vector space over $\mathbb{R}$ for which $\dim(V) = 2$. Let $\mathcal{A} = (v_1, v_2)$ and $\mathcal{B} = (w_1, w_2)$ be two ordered bases of V and let $T : V \rightarrow V$ be a linear operator. Suppose that

$$[T]_{\mathcal{A}}^{\mathcal{A}} = \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix} \quad \text{and} \quad [T]_{\mathcal{A}}^{\mathcal{B}} = \begin{pmatrix} 3 & 2 \\ -1 & 4 \end{pmatrix}$$

- (a) Compute the transition matrices $[id]_{\mathcal{A}}^{\mathcal{B}}$ and $[id]_{\mathcal{B}}^{\mathcal{A}}$.

Hint. We have $[T]_{\mathcal{A}}^{\mathcal{B}} = [T]_{\mathcal{A}}^{\mathcal{A}} \cdot [id]_{\mathcal{A}}^{\mathcal{B}}$.

- (b) Compute $[T]_{\mathcal{B}}^{\mathcal{B}}$.

4. Let V be a finite-dimensional vector space over the field F and let $T, S \in \text{hom}(V, V)$. Let $\mathcal{A}$ and $\mathcal{B}$ be two ordered bases of V . Suppose that the following matrix identity holds:

$$[S]_{\mathcal{A}}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{A}} = [T]_{\mathcal{A}}^{\mathcal{B}} \cdot [S]_{\mathcal{B}}^{\mathcal{A}}.$$

Prove that $S \circ T = T \circ S$.

5. Let V be a vector space over a field F and let $\mathcal{A} = (v_1, \dots, v_n)$ be an ordered basis of V . Let $T : V \rightarrow V$ be a linear operator. Suppose that $\mathcal{B} = (T(v_1), \dots, T(v_n))$ is an ordered basis of V . Prove that

$$[T]_{\mathcal{A}}^{\mathcal{A}} = [T]_{\mathcal{B}}^{\mathcal{B}}$$

6. Prove or disprove each of the following statements:

- (a) The matrices given by

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix}$$

are similar.

- (b) The matrices given by

$$A = \begin{pmatrix} 1+i & 2 \\ i & 1-2i \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 1+i & 2 \\ 2i & 1-i \end{pmatrix}$$

are similar.

- (c) Let $K = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Then the set

$$W = \{A \in M_2(\mathbb{R}) : A, K \text{ are not similar}\}$$

is a subspace of $M_2(\mathbb{R})$.

- (d) Let $\alpha, \beta \neq 0$. Then the matrices $A = \begin{pmatrix} 0 & \alpha \\ 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 0 \\ \beta & 0 \end{pmatrix}$ are similar.